National University of Computer & Emerging Sciences, Lahore

AI2002 – Artificial Intelligence (BS-SE-5A) — Project Deliverable 5: Technical Report (Track B)

Lightweight Speech Emotion Recognition Using Hand-Crafted Acoustic Features: Experimental Results, Theoretical Insights and Reproducibility

Group ID	F26-11	**Track**	B – Research & Development
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Code	F26-11.ipynb (also F26-11.py) · https://github.com/junaidshahid-dev/Ai-project (folder phase5/)		

1. Experimental Results

We re-implemented the lightweight SER ensemble of Chowdhury et al. (*Sci. Rep.* 15, 11824, 2025): 34 hand-crafted features per frame (ZCR, RMS energy, 20 MFCCs, 12 Chroma) for 108 frames, flattened to 3,672 values, fed to a 1D-CNN and a CNN-BiLSTM whose probabilities are averaged. The paper reports 97.57% on RAVDESS. Table 1 summarises our results (details in Deliverable 4).

Protocol	Test data	Ensemble	SVM	Paper
Augment before split (leaky)	576 RAVDESS samples	96.5%	85.6%	97.57%
Clean random split (80/10/10)	144 RAVDESS clips	81.9%	77.1%	–
Unseen speakers (4-fold)	1,440 clips, new actors	58.5% ± 1.9	49.1%	–
RAVDESS → TESS	2,800 TESS clips	19.7%	23.9%	– (chance 14.3%)
Gender gap, unseen speakers	720 clips per gender	F 66.2% / M 50.7%	F 52.9% / M 45.3%	–

Table 1. Accuracy of the paper's ensemble and the SVM baseline under each protocol.

Error analysis in brief. Most errors are between acoustically similar emotions (sad ↔ neutral, happy → fearful, angry → disgust); sad is the hardest class for new speakers (34.4% recall); strong intensity is far easier than normal intensity (66.1% vs 47.9%); accuracy per actor ranges from 40.0% to 81.7%; and on TESS the model predicts *sad* for 53.1% of clips, largely because 41.5% of TESS frames are zero padding. Below we test four explanations with new experiments, using the SVM because it trains in seconds.

2. Reproducibility of the Analysis

- **Every reported number is recomputed.** The notebook recomputes all 30 accuracies of the experiment summary (6 models × 5 protocols) from the saved per-clip predictions: all are identical.
- **Re-training gives the same predictions.** Re-training the SVM from the feature cache with seed 42 on this laptop gives the **same prediction for 100.0%** of the random-split test clips (77.1% accuracy, as saved) and 98.7% of the 1,440 unseen-speaker clips (49.3% vs 49.1% saved). The Random Forest agrees on 89.6% (67.4% vs 69.4%): its trees depend on the library version and the number of CPU threads, and deep models change in the same way (our Phase 3 re-run gave the ensemble 77.1%).
- **Single test splits are noisy (Figure 3, left).** Repeating the random 80/10/10 split with 10 seeds gives the same SVM between **72.9% and 83.3%** (mean 76.9%, std 4.1 points). Theory predicts a standard error of $\sqrt{p(1-p)/n} = 3.5$ points for n = 144 test clips. So a difference of a few points between two models, or between our run and a re-run, is within normal variation; conclusions should rest on large test sets such as the 1,440-clip speaker-independent protocol.

3. Theoretical Insights and Follow-up Experiments

Insight 1 – the features describe the speaker more than the emotion. For each of the 68 summary features (mean and std of the 34 features over time) we computed eta squared, the share of its variance explained by group means. On average the speaker explains 27.9% of a feature's variance and the emotion only 7.6%; the speaker dominates in 65 of the 68 features. The same SVM recognises **who is speaking with 95.0% accuracy** (24 actors, chance 4.2%) but the emotion with 74.4% (5-fold CV). Only RMS energy is mainly driven by emotion (34.2% vs 4.0%). This explains the drop for unseen speakers: a classifier can reach high within-speaker accuracy by learning each speaker's typical feature values, and those values are wrong for a new voice.

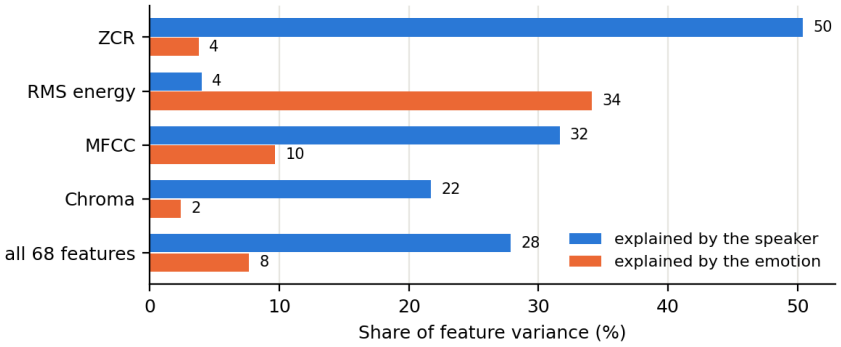


Figure 1. Average share of feature variance explained by the speaker and by the emotion (RAVDESS, 1,440 original clips).

Insight 2 – removing speaker identity helps, but does not remove the gender gap. If speaker identity is the problem, z-scoring each speaker's features with that speaker's own mean and std should help. This needs no labels, so it is also possible for a new user (a short enrolment recording). Results with the SVM (Figure 2):

- Unseen speakers: **49.3%** → **59.2%** (+9.9 points). The simple SVM then matches the paper's deep ensemble without normalisation (58.5%).
- RAVDESS → TESS: **24.2%** → **34.8%**, the best cross-dataset result in the project, because normalisation also removes the differences in recording level and voice between the two corpora.
- Gender gap: 7.5 → **11.0 points**. Both genders improve (male 45.6% → 53.8%, female 53.1% → 64.7%), but women gain more. The bias is therefore *not* caused by different voice baselines; the emotional changes in the male voices of RAVDESS are harder to separate with these features (possibly because of a smaller pitch range, which Chroma and MFCC capture). Fixing it would need other features or gender-balanced model selection.

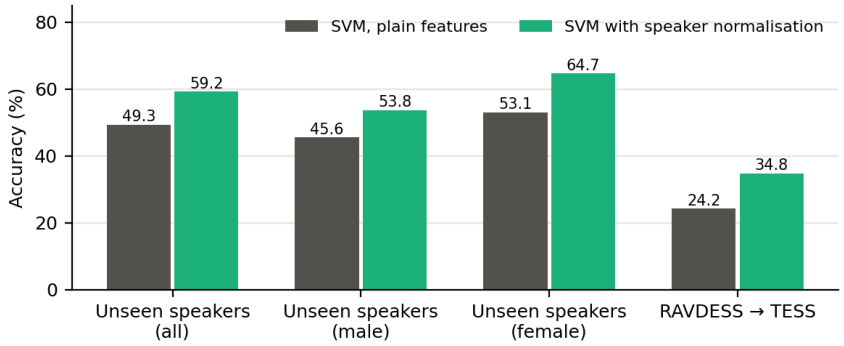


Figure 2. SVM accuracy with and without per-speaker normalisation.

Insight 3 – the flattened input is tied to time positions. The paper's 3,672-value vector gives each frame its own position, so the same speech starting a little later produces a different input. We trained an SVM on the flattened input and one on the 68-value summary, and tested both on test clips delayed by 0–40 frames (Figure 3, right). A delay of only 0.12 s (5 frames) costs the flattened input 15.3 points (67.4% → 52.1%) but the summary only 2.1 (77.8% → 75.7%); at 0.23 s the losses are 26.4 and 9.0 points. This explains why the Random Forest, which uses the summary, transfers best to TESS (31.1%), and why the zero padding of short TESS clips hurts the deep models: loading TESS from 0 s instead of 0.6 s raised the ensemble from 19.7% to 30.5%. Convolutions detect a pattern wherever it occurs, but the Flatten and Dense layers after them again assign weights to fixed positions.

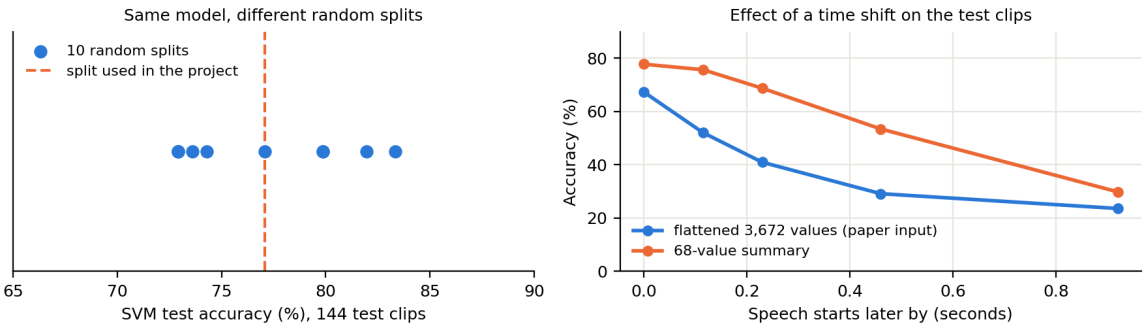


Figure 3. Left: SVM accuracy on 10 random splits (dashed: the split used in the project). Right: accuracy when the test speech is delayed.

Insight 4 – why augmentation before splitting inflates accuracy. A noisy or pitch-shifted copy of a recording is almost the same input as the original. When copies are split at random, 99.1% of test samples have a copy in training, so a high-capacity model can recognise the recording instead of the emotion. The test set is then no longer independent of the training set, which every accuracy estimate assumes. This is the only setting in which we reach the paper's level (96.5%).

4. Conclusions

- The paper's architecture works (81.9% on a clean split, best model for unseen speakers), but its 97.57% is only reproduced with leakage between training and test data.
- The main weakness is speaker dependence: the features carry more information about the speaker than about the emotion. Per-speaker normalisation is a cheap, label-free fix (+9.9 points for new speakers, +10.6 on TESS with an SVM) and should be applied to the deep models next.
- The gender gap is a separate problem that normalisation does not solve; it must be reported and checked before any deployment.
- Results must come with their uncertainty: on 144 test clips, the split alone moves accuracy by up to 10 points.

5. Challenges Faced

- **Testing explanations cheaply.** Re-training the deep models takes hours on our 2-core CPU, so the follow-up experiments use the SVM on the same features. It follows the same trends as the deep models (good on a random split, much weaker for new speakers, near chance on TESS), but the effect of normalisation on the deep ensemble itself is still to be measured.
- **Exact reproducibility.** Seeds alone are not enough: Random Forest and neural-network results change with the library version and number of CPU threads. We save all predictions and report how many of them a re-run reproduces.
- **A surprising result.** We expected speaker normalisation to reduce the gender gap; it increased it. We report it as it is, because it shows that the bias has a different cause.

To reproduce: run `phase5/F26-11.ipynb` (about 6 minutes on a laptop CPU; needs the Phase 2 feature cache, no TensorFlow). All seeds are 42.